

The Special Theory of Relativity and the Lorentz Transformation

1 Introduction

The Special Theory of Relativity is a physical theory regarding the relationship between space and time. It is based on two fundamental postulates:

1. The laws of physics are invariant in all inertial frames of reference.
2. The speed of light in a vacuum, c , is the same for all observers, regardless of the motion of the light source or observer.

2 The Spacetime Interval

A primary consequence of these postulates is that the distance between two events in spacetime, known as the spacetime interval ds^2 , must be invariant under a change of inertial frames. In flat Minkowski space, this interval is defined as:

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2 \quad (1)$$

For any two observers in different inertial frames, the condition $ds^2 = ds'^2$ must hold.

3 The Lorentz Transformation: A Solution

The equations that satisfy the invariance of the spacetime interval are the Lorentz transformations. For a reference frame S' moving with a constant velocity v along the x -axis relative to a frame S , the coordinates (t, x, y, z) and (t', x', y', z') are related by:

$$t' = \gamma \left(t - \frac{vx}{c^2} \right) \quad (2)$$

$$x' = \gamma(x - vt) \quad (3)$$

$$y' = y \quad (4)$$

$$z' = z \quad (5)$$

where the Lorentz factor γ is defined as:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (6)$$

These transformations represent the mathematical solution to the requirement that the speed of light remains constant across all inertial frames.

4 Relativistic Effects

The Lorentz transformations lead to several significant physical predictions:

- **Time Dilation:** A clock moving relative to an observer will be measured to tick slower than a clock at rest in the observer's own frame, expressed as $\Delta t = \gamma \Delta t_0$.
- **Length Contraction:** The length of an object in motion is measured to be shorter than its proper length in the direction of motion, expressed as $L = \frac{L_0}{\gamma}$.
- **Mass-Energy Equivalence:** The theory unifies mass and energy through the relation $E^2 = (pc)^2 + (m_0c^2)^2$, which simplifies to $E = mc^2$ for an object at rest.